

Caden Cheng

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EDUCATION

University of California, San Diego

B.S. in Computer Science and Mathematics, Minor in Data Science

La Jolla, CA

Expected Grad 2028

GPA: 3.675 | **Relevant Coursework:** Object-Oriented Programming (Java), Discrete Math, Linear Algebra

Piedmont Hills High School

Weighted GPA: 4.05

San Jose, CA

Aug. 2021 – June 2025

EXPERIENCE

Software Developer

Computer Science Engineering Society (CSES) at UCSD

October 2025 – Present

La Jolla, CA

- Developed a full-stack portfolio application using **React**, **Next.js**, and **TypeScript**, serving dynamic content via **REST APIs**
- Implemented industry-standard **Git** workflows, using feature branching and **Pull Requests** for code integration
- Engineered responsive UI components with **Tailwind CSS** and **React hooks** to ensure cross-device compatibility
- Architected a **MongoDB** database using **Mongoose** schemas to validate and structure project data collections

High School Math Tutor

Piedmont Hills High School

Sept. 2024 – May 2025

San Jose, CA

- Mentored **50+ students** to improve academic performance in Algebra, Calculus, and Statistics curricula
- Designed customized lesson plans to prepare students for standardized exams (**SAT/ACT**, **AP Calculus**)
- Simplified complex mathematical concepts into digestible steps, adapting strategies for diverse learning styles

PROJECTS

Silent Speech | *PyTorch, OpenCV, TensorFlow, Python*

January 2026

- Best Overall Hack** at San D Hacks 2026 (out of 35+ teams)
- Developed a real-time lip-reading AI using a **dual-stream BiGRU** and **Transformer encoder**, achieving high-accuracy video-to-text transcription on live webcam feeds
- Engineered a robust pipeline using **OpenCV** for facial landmark extraction, normalizing features against mouth width to ensure scale invariance across different users
- Optimized training using stratified sampling and data augmentation (mixup, noise injection) to prevent overfitting

ACM Blockography AI Competition | *Python, PyTorch, Computer Vision*

November 2025

- Fine-tuned a pretrained **ResNet-50 convolutional neural network** to detect **Minecraft biomes** from game imagery
- Implemented an end-to-end machine learning pipeline including data preprocessing, augmentation, and validation
- Optimized model performance using early stopping and learning-rate scheduling techniques
- Achieved **90% validation accuracy** on biome classification dataset, automating inference for competition submissions

NASA Exoplanet Classification AI | *Python, TensorFlow, Pandas*

June 2024 – August 2024

- Developed a deep learning classifier to detect exoplanets using time-series transit photometry data from NASA
- Engineered a data preprocessing pipeline using **Fourier Transforms** to reduce signal noise in light curves
- Trained a **1D Convolutional Neural Network (CNN)** using TensorFlow/Keras, achieving **85% accuracy**
- Fixed dataset imbalance using SMOTE to generate more training examples

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks: React, Next.js, Node.js, Tailwind CSS, PyTorch, TensorFlow

Developer Tools: Git/GitHub, MongoDB Atlas, npm

Libraries: Pandas, NumPy, Matplotlib, Torchvision, OpenCV